

Iterative Prompting

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of controlled prompt experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

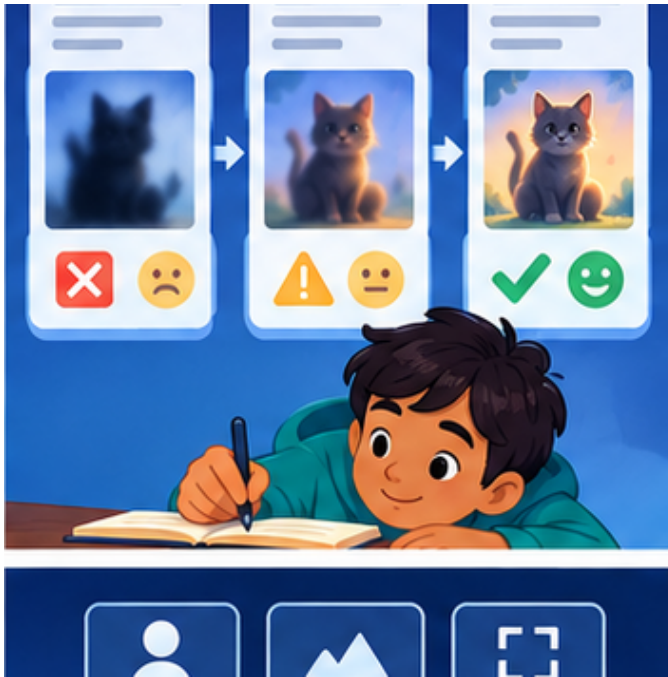
Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 7: Iterative Prompting

Revision should respond to diagnosed output failures.



CREATOR MISSION

Keep a three-version prompt log with reasons for each change.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Structured outputs and validation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

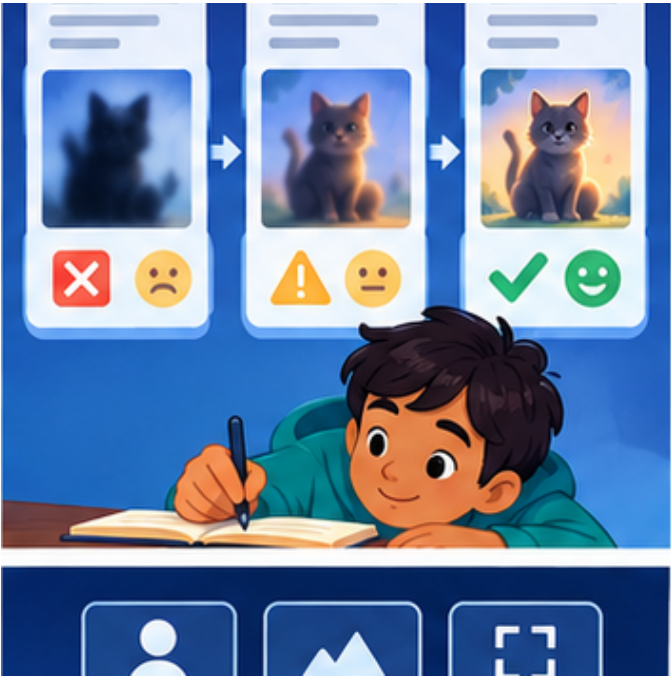
By the end, I can revise one prompt component and compare paired outputs using the same rubric.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use iterative prompting to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting outputs; change one intended instruction and preserve other settings.

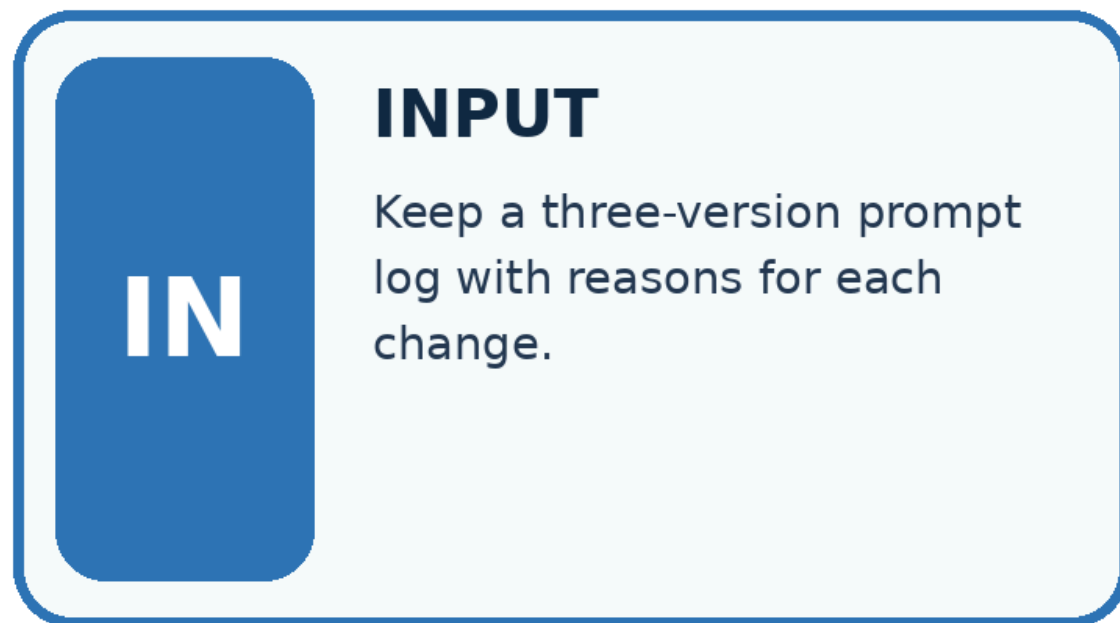
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Revision should respond to diagnosed output failures.



OUT

EVIDENCE

Keep a three-version prompt log with reasons for each change.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric. Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Controlled prompt experiments — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Check or calculation: Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.

Conclusion and limit: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Change model settings and note wording at the same time as the prompt.

Prediction: The comparison no longer isolates the prompt revision.

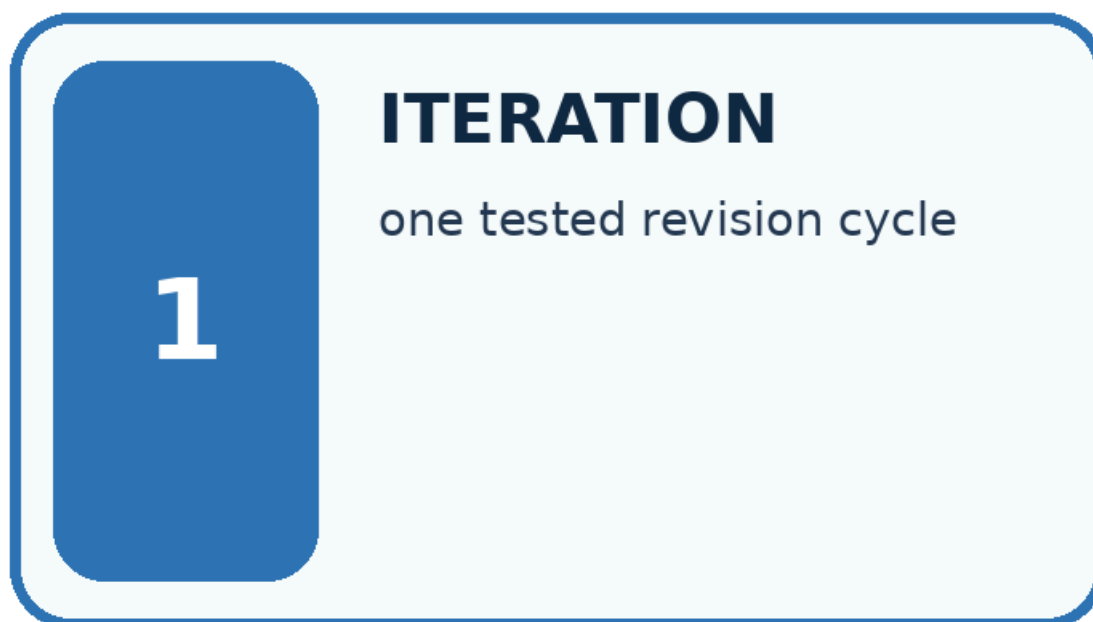
Reason: Several changes could explain the altered outputs; record them or run separate controlled comparisons.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

FAILURE DIAGNOSIS

specific reason an output
missed the goal

3

VERSION

saved prompt state

4

CHANGE LOG

what changed and why

Word · My own example · Not this / boundary

Word: iteration

My own example: One repeat of a process, often followed by inspection and revision.

Not this / boundary: _____

Word: failure diagnosis

My own example: An investigation locating where and why a process failed.

Not this / boundary: _____

Word: version

My own example: An identified state of a document, dataset, model or program.

Not this / boundary: _____

Word: change log

My own example: A record of what changed, when it changed and why.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Paired case

Meaning: The same case evaluated under each compared procedure

Word: Rubric

Meaning: Prespecified criteria for judging an output

Word: Regression

Meaning: A previously passing condition that fails after a change

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 7 • WORKED EVIDENCE

- 1 Version 1 fails: answer is too advanced.
- 2 Change only audience constraint; Version 2 is age-fit but lacks evidence.
- 3 Add source-use rule; Version 3 cites the supplied passage.
- 4 Record scores and reasons so improvement is traceable.

1. Version 1 fails: answer is too advanced.
2. Change only audience constraint; Version 2 is age-fit but lacks evidence.
3. Add source-use rule; Version 3 cites the supplied passage.
4. Record scores and reasons so improvement is traceable.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Iterative Prompting

A fictional Study Buddy prompt invents a definition when approved notes omit it.

The facts we will use

P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

Iterative Prompting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate failure

The target defect is unsupported answers when notes are missing.



2. One controlled change

Q adds the missing-evidence rule; other instructions remain fixed.



3. Paired results

P=2/4; Q=3/4; gain is case 3, with case 4 unresolved.



4. Next action

Inspect case 4 and reserve genuinely fresh questions for final evaluation.

CHANGE ONE THING

Change model settings and note wording at the same time as the prompt.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Locate failure

What happens and why: The target defect is unsupported answers when notes are missing.

Step: 2. One controlled change

What happens and why: Q adds the missing-evidence rule; other instructions remain fixed.

Step: 3. Paired results

What happens and why: P=2/4; Q=3/4; gain is case 3, with case 4 unresolved.

Step: 4. Next action

What happens and why: Inspect case 4 and reserve genuinely fresh questions for final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

An iteration should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too. The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

What this example does not establish

Stochastic output can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.

Read the labelled picture: What evidence supports improvement here, and what remains unproven?

Change one thing: Change model settings and note wording at the same time as the prompt. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Keep adding instructions until an output looks better. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Prompt selection on development cases cannot be reported as independent final performance.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled prompt experiments”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What evidence supports improvement here, and what remains unproven?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Use a fixed prompt and test set. Diagnose one failure, change one element, score again, and preserve all three versions.

Use a fixed prompt and test set. Diagnose one failure, change one element, score again, and preserve all three versions.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

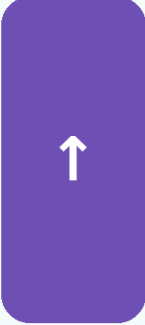
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Maintain prompt version, diagnosed failure, change, and before/after rubric scores.

THIS WEEK’S CHECKPOINT Maintain prompt version, diagnosed failure, change, and before/after rubric scores.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Maintain a Study Buddy experiment log linking failure, single revision and paired rubric results.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Keep a three-version prompt log with reasons for each change.

CHALLENGE Keep a three-version prompt log with reasons for each change.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 7 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A factually correct explanation is too difficult for a nine-year-old.

The question

Make and test one focused revision.

Worked explanation

Constrain vocabulary and request a concrete example. Compare readability and factual preservation on the same passages.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Revise one cause and retest

Evidence or stage · Value or result

Evidence or stage: Diagnose

Value or result: Too difficult for a nine-year-old

Evidence or stage: Revise

Value or result: Simpler vocabulary and one concrete example

Evidence or stage: Retest

Value or result: Use the same passages

Evidence or stage: Compare

Value or result: Readability and factual preservation

A factually correct explanation is too difficult for a nine-year-old.

Make and test one focused revision.

Constrain vocabulary and request a concrete example. Compare readability and factual preservation on the same passages.

14. Try a new case independently

Independent case: Changed independent case

A factual response uses four paragraphs when the contract requires three bullets. Design a controlled prompt revision and evaluation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 7 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You change a prompt's audience, length and source rules together; the result improves. What can you conclude, and how would you investigate the cause?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Maintain prompt version, diagnosed failure, change, and before/after rubric scores.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional Study Buddy prompt invents a definition when approved notes omit it.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of controlled prompt experiments

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of controlled prompt experiments

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy prompt invents a definition when approved notes omit it.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect case 4 and reserve genuinely fresh questions for final evaluation.

- 1. Locate failure
- 2. Next action
- 3. One controlled change
- 4. Paired results

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Next action Inspect case 4 and reserve genuinely fresh questions for final evaluation. An iteration should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too. The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

Practice 2: explain the deciding evidence

Changed-case practice: Change model settings and note wording at the same time as the prompt. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. The comparison no longer isolates the prompt revision.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The comparison no longer isolates the prompt revision. Several changes could explain the altered outputs; record them or run separate controlled comparisons.

Repair a missing building block

Review Examples and Rubrics (ai-3/6). Goal: Write a prompt specification with observable response criteria. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?. Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative prediction from generation by inspecting the requested output. Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.